

Python Lesson 2

Miss Wallace

"Talking to Computers"

Python Quiz

- You have 15 minutes to work on this in your tables! Write your answers to each question and – very important! - write your ID codes on the post-it on your table.

Starter

Question: can you have a conversation with a computer?

Have a think about this during the lesson! I'll ask for your thoughts at the end.

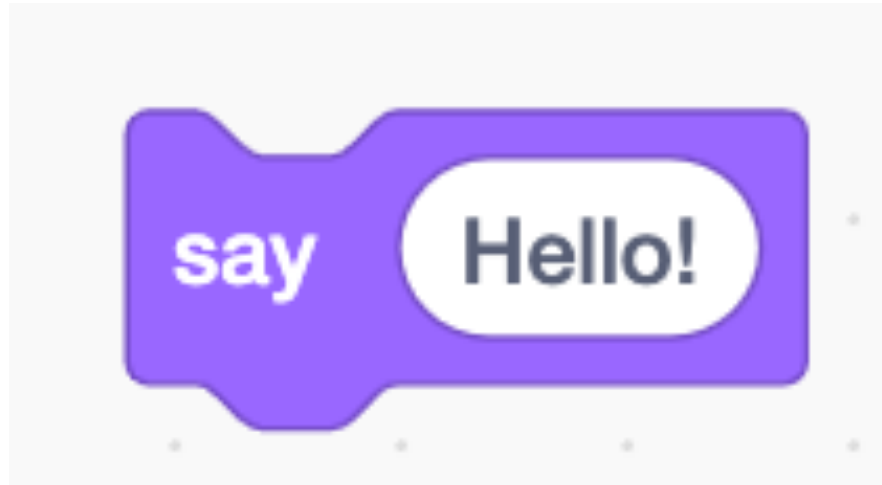


Image credit: screenshot from
<https://scratch.mit.edu/projects/editor/?tutorial=getStarted>

Conversations

- Turn around to the person next to you. I want you to imagine you have never met before (or that your memory of them has been completely wiped...)
- You want to get to know them so you need to ask questions! You should each ask the other person a question and take turns asking these questions, just like you would in a regular conversation.
- Write down on your **reflection sheets** your mini-conversation where you each ask a question. Make sure you also write down your answers and anything else you might say!

What conversations have you had?

- I'm going to write up on the board some examples of conversations you have all been having
- Are there any patterns you can spot?
- Do you think we could model this using code?
- Let's try!

Your task

Your task is to model your conversation using Python

We're going to do this in stages

Pick just **one** of your questions. Make sure you have an answer and response to it written down!

- 1) Pick three different colours. Look at your question and mark with a different colour each of the following conversation constructs we have identified: **statement**, **question** and **response**.

Conversations in Python

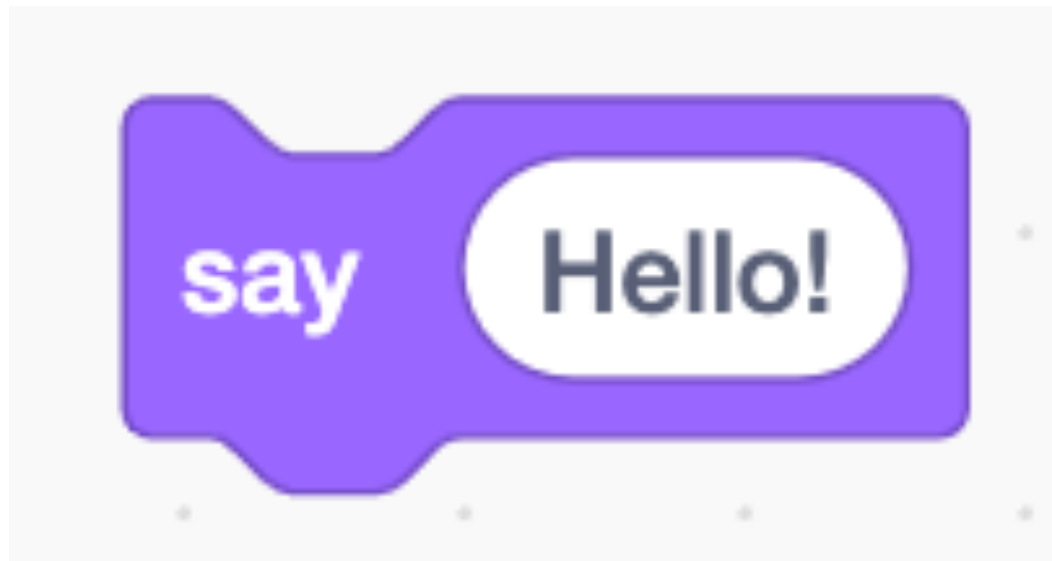
Let's have a look at this code. What words/phrases do we recognise? Do we have any guesses at what this code might do?

```
1  input("What's your name?")  
2  print("Nice to meet you!")
```

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Conversations in Python

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```

Let's see what happens!

Conversations in Python

```
1  input("What's your name?")  
2  print("Nice to meet you!")
```

Does this look like having a conversation? Why or why not? Does anything change depending on what you say your name is?

Your task

Your task is to model your conversation using Python

We're going to do this in stages

Pick just **one** of your questions. Make sure you have an answer and response to it written down!

- 1) Pick three different colours. Look at your question and mark with a different colour each of the following conversation constructs we have identified: **statement**, **question** and **response**.
- 2) Let's think about how we map these the Python we have seen so far. Write down on your **reflection sheets** your "translation" into Python.

Your task

Your task is to model your conversation using Python

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Pick just **one** of your questions. Make sure you have an answer and response to it written down!

- 1) Pick three different colours. Look at your question and mark with a different colour each of the following conversation constructs we have identified: **statement**, **question** and **response**.
- 2) Let's think about how we map these the Python we have seen so far. Write down on your **reflection sheets** your "translation" into Python.
- 3) Implement this in your own program. When you do this, things might go wrong... That's absolutely fine! Let's think about how we might deal with this.

Python Coding + Debugging

- When things in our code don't work, we need to try and fix this. We call this process **debugging**.
- When things go wrong, you will get **error messages** on screen. These look like this:

```
SyntaxError: unterminated string literal (detected at line 1)
on line 1 of main.py:
name = input(What's your name?)
            ^
```

```
NameError: name 'inpu' is not defined. Did you mean: 'input'?
on line 1 of main.py
```

```
IndentationError: unexpected indent on line 2 of main.py
```

Debugging

- Tell me what you think has gone wrong in each of the following examples!

Python Coding + Debugging

- Go to the Python editor!
- Try and type in your conversation program.
- If something goes wrong, try and fix it **but make sure you write down what you had to change! It's really helpful for me to see this so I get an idea of how you are all thinking about these problems 😊**

Conversation programs

- How did we get on? What did we find difficult?
- Does it feel like a “real” conversation? Why or why not?

Real life input-output processes

Lots of programs ask for your input!

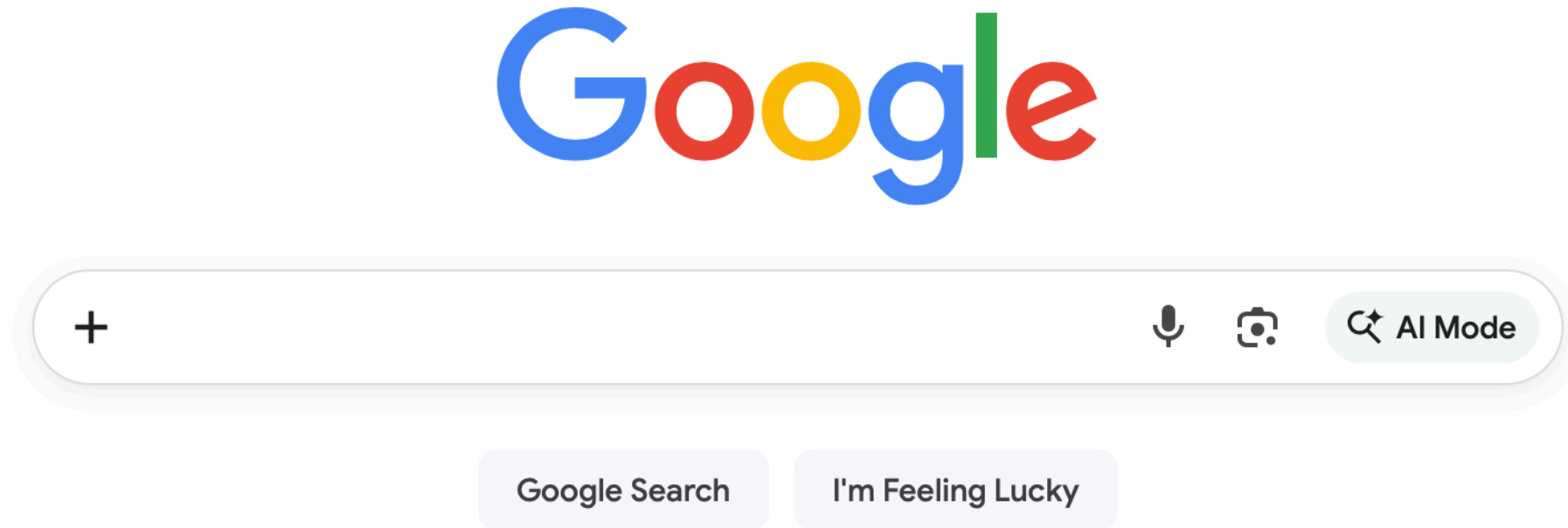


Image credit: screenshot of
https://www.google.co.uk/?client=safari&channel=iphone_bm

Real life input-output processes

Lots of programs ask for your input!

A screenshot of the Amazon.co.uk search bar. It features a dark blue header bar. On the left, there is a light grey button with the text 'All' and a downward-pointing triangle. To the right of this button is a large white search input field with the placeholder text 'Search Amazon.co.uk'. On the far right of the search bar is an orange square button containing a white magnifying glass icon.

Can anyone think of any other examples of this?

Exit passes and next time...

- Please fill out your exit passes
- I was really impressed with the detail in your answers from lesson 1 so thank you all very much 😊 Any comments and feedback you have for me are very useful and much appreciated!